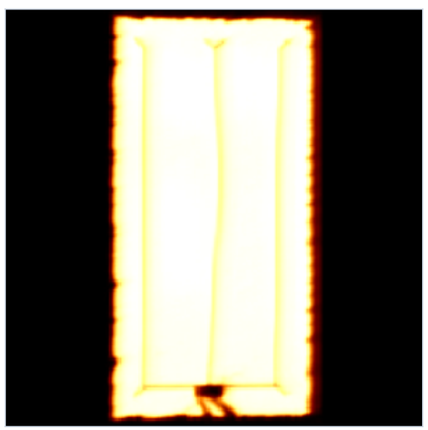
CIGS 0279 -- 20% sampling -- stored noisy measurement

NPL-default PCM noise | Normalised Hadamard J=8, coarse J=6 | AFM-hot, fixed intensity [0,1]

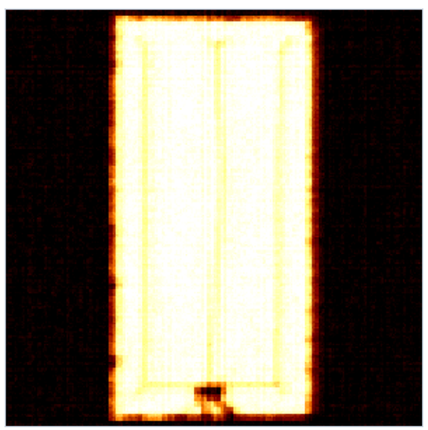
Target



PSNR inf dB | SSIM 1.0000

HaarPSI 1.0000 | MSE 0.000e+00

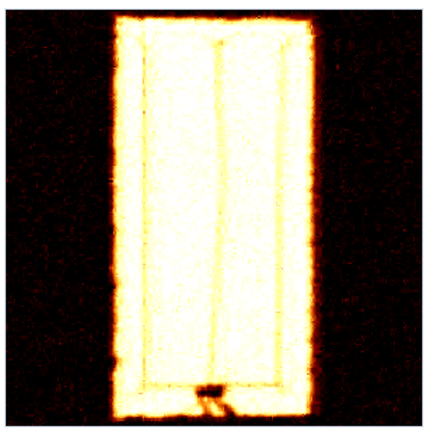
Zero-filled



PSNR 26.51 dB | SSIM 0.5039

HaarPSI 0.6453 | MSE 2.232e-03

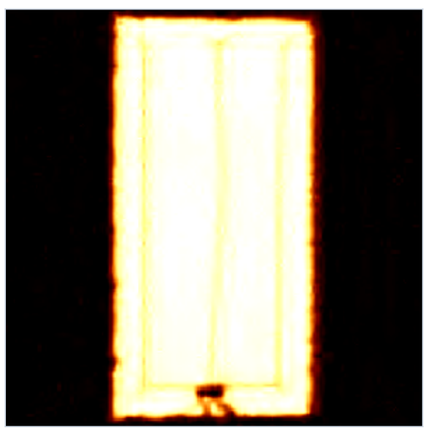
SPGL1-BP



PSNR 27.01 dB | SSIM 0.4674

HaarPSI 0.7333 | MSE 1.989e-03

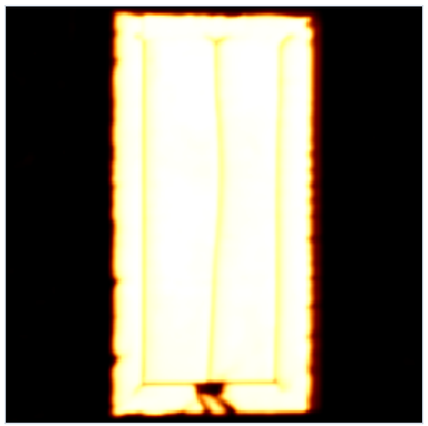
SPGL1-BPDN



PSNR 32.31 dB | SSIM 0.8341

HaarPSI 0.7670 | MSE 5.881e-04

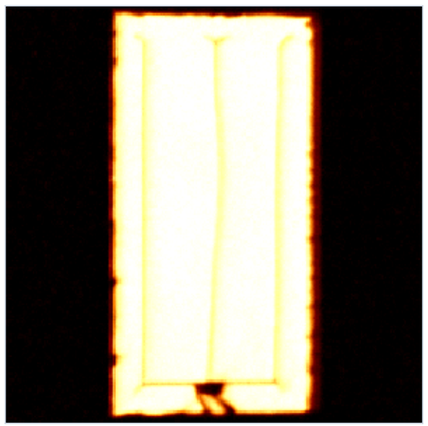
PnP-ADMM-DRUNet-tuned



PSNR 40.27 dB | SSIM 0.9877

HaarPSI 0.9597 | MSE 9.392e-05

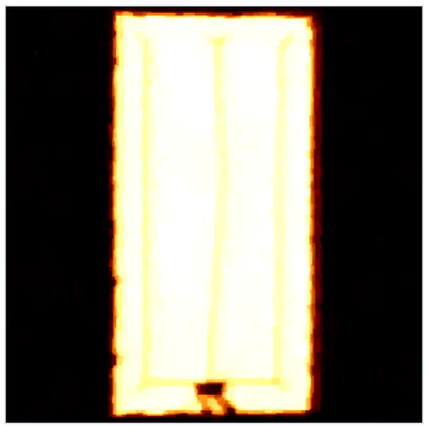
GS-PnP-GS-DRUNet-tuned



PSNR 34.72 dB | SSIM 0.8348

HaarPSI 0.9261 | MSE 3.375e-04

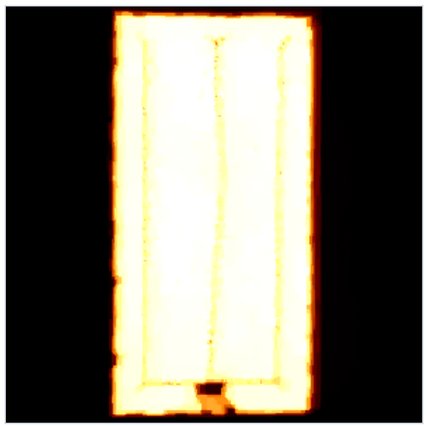
TV-best scalar



PSNR 33.51 dB | SSIM 0.9173

HaarPSI 0.8294 | MSE 4.453e-04

UTV C60 (original)



PSNR 33.42 dB | SSIM 0.9372

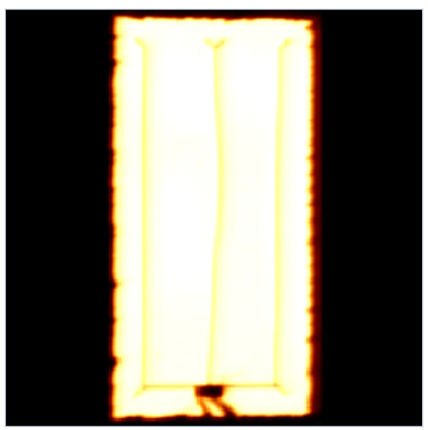
HaarPSI 0.8287 | MSE 4.548e-04

All methods use one stored 20% acquisition. TV-best scalar uses a test-image oracle weight (lambda=0.0322552321) at 512 iterations.

CIGS 0279 -- 10% sampling -- newly simulated noisy measurement

NPL-default PCM noise | Normalised Hadamard J=8, coarse J=6 | AFM-hot, fixed intensity [0,1]

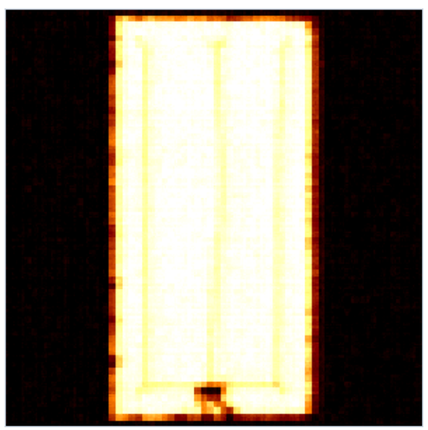
Target



PSNR inf dB | SSIM 1.0000

HaarPSI 1.0000 | MSE 0.000e+00

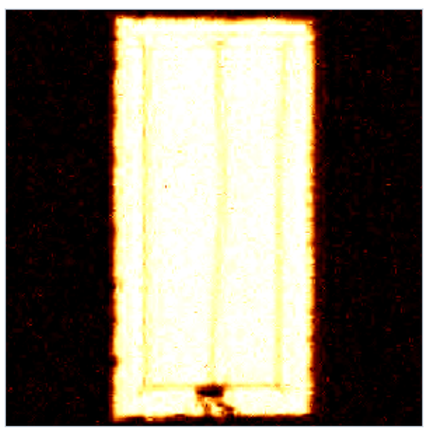
Zero-filled



PSNR 26.19 dB | SSIM 0.6104

HaarPSI 0.5228 | MSE 2.402e-03

SPGL1-BP



PSNR 27.04 dB | SSIM 0.5165

HaarPSI 0.7089 | MSE 1.976e-03

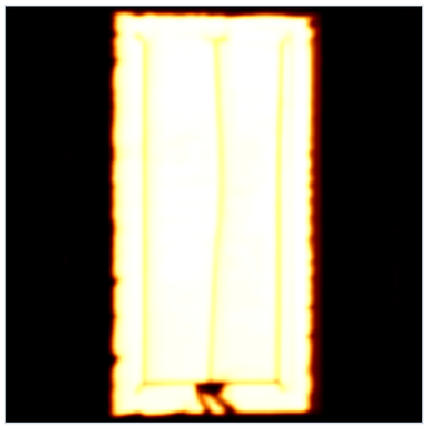
SPGL1-BPDN



PSNR 30.72 dB | SSIM 0.8084

HaarPSI 0.7207 | MSE 8.473e-04

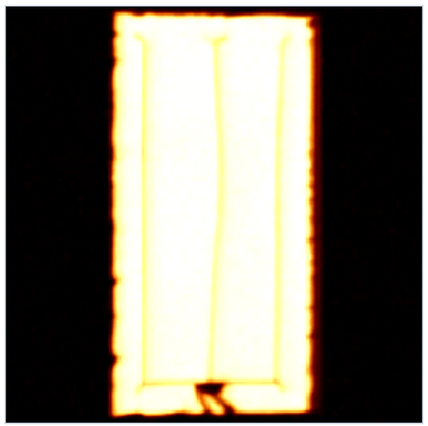
PnP-ADMM-DRUNet-tuned



PSNR 38.09 dB | SSIM 0.9804

HaarPSI 0.9343 | MSE 1.553e-04

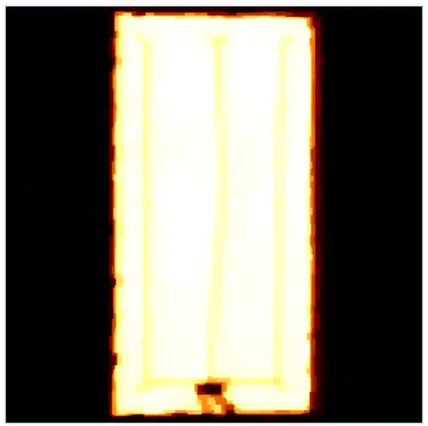
GS-PnP-GS-DRUNet-tuned



PSNR 37.81 dB | SSIM 0.9163

HaarPSI 0.9293 | MSE 1.656e-04

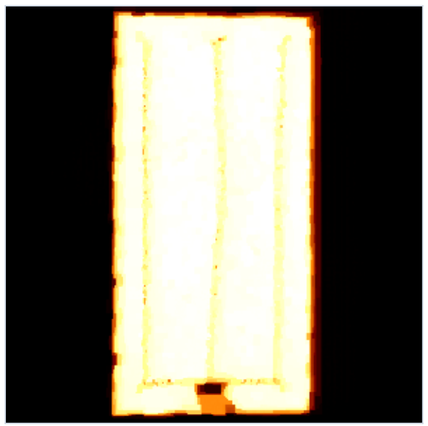
TV-best scalar



PSNR 31.99 dB | SSIM 0.9085

HaarPSI 0.7386 | MSE 6.324e-04

UTV C60 (original)



PSNR 31.26 dB | SSIM 0.9216

HaarPSI 0.7642 | MSE 7.479e-04

PnP-ADMM-DRUNet-tuned, GS-PnP-GS-DRUNet-tuned, and TV-best scalar use their 20%-selected parameters and were not re-tuned for 10% sampling.